Amr Abdallah

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Solutions Architect and technical lead with 10+ years of hands-on engineering. Designs end-to-end systems that connect hardware, data, and people: real-time IoT telemetry on Azure, secured APIs over enterprise quality and supply-chain data, and a suite of internal applications behind a single authentication gateway. Owns delivery from requirements and architecture through automated deployment, leading cross-functional teams and contributing directly to implementation.

Core Competencies

- **Solutions Architecture**: Microservices, distributed and event-driven systems, real-time IoT (MQTT, Modbus, WebSocket), secured API design, cloud infrastructure design.
- **Azure**: Functions, App Service, Container Apps, Virtual Machines, Cosmos DB, Blob Storage, SignalR, API Management, Entra ID, Azure Data Explorer, Power BI.
- **AWS**: Lambda, API Gateway, EC2, Elastic Beanstalk, container services, Amplify, DynamoDB, MemoryDB, S3, EFS, IAM, CloudWatch, CloudTrail, Network Firewall, AWS Backup.
- **DevOps & CI/CD**: Docker, Terraform, CI/CD pipeline design with Jenkins and GitHub webhooks, Gitea self-hosted on the internal network; automated testing with Jest, Cypress, and Playwright.
- **Full-Stack Engineering**: MERN (MongoDB, Express, React, Node.js), Go, Python (FastAPI/Flask), Java (Spring), .NET Core/C#, PHP (Laravel), Vue.js, TypeScript, REST APIs, GraphQL.
- **Data Systems**: PostgreSQL, MySQL, Oracle, DB2, MS SQL, MongoDB, Kusto, Grafana, Redis, Supabase, vector databases.
- **AI & Agentic Systems**: AI agents, LLM integration, retrieval-augmented generation (RAG), tool/function calling, Model Context Protocol (MCP), prompt engineering, Microsoft Copilot Studio, multi-agent coding with Claude Code, local LLMs, predictive modeling.
- **Technical Leadership & Delivery**: Full-lifecycle delivery, requirements, specifications and Visio diagrams, backlog prioritization, Agile/Scrum (Jira, Azure DevOps), cross-functional team leadership.

Professional Experience

- **AlumaPower**
  Senior Software Engineering Lead *2022 - Present*
  - Engineered a real-time IoT data pipeline: hardware boards publish MQTT messages to Azure Web Apps that decode and normalize multi-sensor readings onto shared timestamps, feeding Kusto (Grafana dashboards), MongoDB, and a live client that charts telemetry and starts/stops fans and motors
  - Built the company's analytics platform, turning operational logs and test-technician experiment data into the KPIs behind company decisions and reports, with Docker-hosted Redis caching large queries
  - Architected a company-wide MERN suite behind one authentication gateway, spanning QMS, SCM, HR, assets, quality data, fabrication, and ticketing, saving thousands of hours
  - Delivered a Copilot Studio agent running on Claude, used company-wide in Teams (thousands of questions a month), that answers quality, purchasing, and KPI questions and flags recurring failures
  - Implemented the secured, read-only Azure API behind the agent (Entra ID), exposing curated MongoDB views as LLM tools that match QMS records with NCRs and ECRs
  - Set up CI/CD with Jenkins and GitHub webhooks for all web apps, Terraform provisioning, and a self-hosted Gitea server on the internal network; own unit, regression, and end-to-end testing
  - Lead the engineering team, owning hiring, performance reviews, and delivery with leadership and partner teams
  - Developed a C# WinForms data logger and controller that charts live voltage, current, temperature, and flow and actuates hardware; now migrating it to React and .NET Core
  - Created the data-cleaning pipeline feeding Monolith AI's platform, whose predictive models for experiments and chemical reactions replaced many physical tests and cut costs by 32%

- **Industrial Web Apps**
  IT Project Manager *2021 - 2022*
  - Contributed to SHORE, the company's end-to-end project management platform, built with React, Node.js, and PostgreSQL on AWS (API Gateway, EC2, Elastic Beanstalk, S3, IAM, CloudTrail)
  - Directed cross-functional teams across the full project lifecycle, maintained the backlog, and prioritized work across simultaneous development and deployment projects
  - Set up Docker-based development environments and web deployments
- **Lambton College**
  Professor (Part-Time) *2021 - Present*
  - Teach Front-End Web Development I (HTML5, CSS3) and II (JavaScript, jQuery, Vue, React), and supervise co-op students
- **Eye-Look Media**
  Full-Stack Developer *2020 - 2021*
  - Authored most of the Laravel backend and the Electron content editor of a digital signage portal (Laravel, Vue, MySQL) for clients such as Canadian Tire
  - Developed a client-facing progressive web app over WebSocket, plus Node.js microservices for automation
- **Avaros EMR**
  Software Developer (Co-op) *2020*
  - Created a patient online-booking frontend (React, React Hooks, RxJS, TypeScript)
  - Integrated Twilio SMS webhooks into a Go backend with PostgreSQL migrations
- **Lambton College Applied Research**
  Researcher (Contracts) *2019 - 2023*
  - Engineered an IoT solution for Just Vertical: a Flutter app over WebSocket to an AWS Lambda microservice, bridging to a Quasar/Electron (Vue.js) app on Raspberry Pi units for irrigation/lighting automation
  - Delivered a MERN-stack operations system for Procor (railcar operations) and a Raspberry Pi digital signage microservice solution for Eye-Look Media
- **Earlier Professional Experience**
  Software Developer and Team Lead, then ICT Teacher
  - Delivered an SMS-based banking solution for a national bank and a Pocket PC sales-force automation system with MS SQL replication; later designed ICT curricula and school software, including a grading system

Selected Projects

- **AI Painting Estimator** (GitHub): Designed and shipped a progressive web app that turns a photo and job description into a professional painting quote. A vision model (Gemini Flash, behind a swappable adapter) flags surfaces, condition, and prep work; a deterministic pricing engine does the arithmetic, and the contractor confirms every price. React, TypeScript, Hono on Cloudflare Workers, D1, R2.
- **Engineering Work Dashboard** (GitHub): Developed an org-wide dashboard that measures engineering output across every repository and branch of a GitHub organization, filtering generated-file noise that can double raw line counts, categorizing commits by type of work, and exporting PDF reports for leadership. Rate-limit-aware, resumable scanning over the GitHub API. Next.js, TypeScript, PostgreSQL (Drizzle), Docker.

Education

- **Lambton College**
  Diploma, Computer Software & Database Development *2019 - 2021*
  Cumulative GPA 3.89/4.0.
- **6th of October University**
  Bachelor of Business Administration, Finance (minor: Economics)
  Degree: Very Good. Graduation Project: Distinct.